Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **event-driven data pipelines**, **TypeScript/Node.js and Python services**, microservices, optimized databases, caching, and observable cloud workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant asynchronous orchestration** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load.
- Achieved 99%+ reliability across 5K+ monthly jobs through **incremental data pipelines**, implementing validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably at scale.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive customer workflows during concurrent model execution and production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows reliably daily.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **microservice platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing role-based access, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, coupon validation, payment webhooks, and Firebase status synchronization while using transactions and locking to prevent conflicting reservations.
- Reduced manual refund handling by approximately 70% through **event-driven payment automation**, supporting refunds, disputes, webhook processing, and consistent status while presenting architecture and backend integration workflows directly to clients reliably.

Skills

Languages: TypeScript, Node.js, Python, JavaScript, SQL, Java, C++
Data & Services: Event-Driven Architecture, Asynchronous Processing, Data Pipelines, Batch Processing, Real-Time Processing, Microservices, REST APIs, WebSockets
AWS & Infrastructure: AWS S3, AWS RDS, AWS Glue, Docker, Kubernetes, Jenkins, CI/CD, Cloud Infrastructure
Databases & Search: PostgreSQL, MySQL, Redis, MongoDB, Azure SQL, Firebase, FAISS, Pinecone, Indexing, Caching, Schema Design
Quality & Operations: Git, GitHub, Unit Testing, Integration Testing, PyTest, Mockito, Observability, Production Debugging, Error Handling, Performance Optimization

Projects

Event-Driven Intelligence Pipeline - Design Concept

- Designed an **AWS event-driven ingestion** concept using EventBridge, SQS, S3, Python workers, retries, dead-letter queues, idempotency, structured metrics, and traceable processing states for resilient high-volume intelligence workflows.

Real-Time Alert Search Service - Design Concept

- Designed a **TypeScript and Node.js alert service** concept with PostgreSQL, Redis caching, OpenSearch indexing, async delivery, query optimization, health metrics, dashboards, and failure diagnostics for low-latency retrieval.

Big-Brain | Live

- Built a **retrieval-augmented generation platform** using LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, and REST APIs, sustaining 95% query accuracy while enabling real-time document analysis reliably for 200+ users.